Set 1D: From a Bode Plot to $H(s)$

ECEN 325 — Lecture 1 (Linear Network Analysis) practice set

What this set trains. Reading an asymptotic (straight-line) magnitude Bode plot backwards: from the corner frequencies and slopes, write $H(s)$ factor by factor; use the one marked level to find the gain K; then compute the steady-state output for a sinusoidal input. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The reverse pipeline: magnitude Bode plot → $H(s)$ (use it for every problem in this set)

Assumptions (as in class): a magnitude plot cannot show the sign of K or whether a zero is in the right half-plane, so take $K > 0$ and all poles/zeros in the left half-plane.
Tools: T5 (sinusoid through H) and T6 (Bode rules) from the Lecture-1 card, plus: a factor $\frac{s}{s+p}$ is $+20\,\text{dB/dec}$ below p and $0\,\text{dB}$ above; $\frac{1}{1+s/p}$ is $0\,\text{dB}$ below and $-20\,\text{dB/dec}$ above; $\left(1+\frac{s}{z}\right)$ is $0\,\text{dB}$ below and $+20\,\text{dB/dec}$ above; dB values of factors add.
Steps:

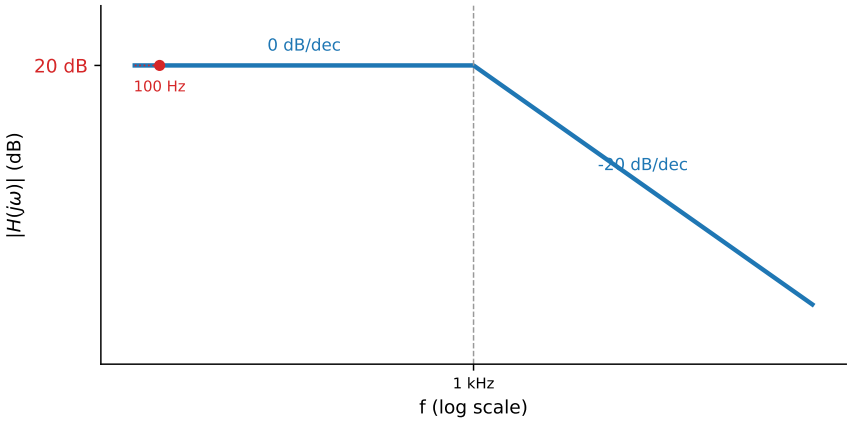
1. *Read the plot:* slope of every segment; at each corner, change $= -20k$ means k poles, $+20k$ means k zeros.
2. *Starting slope* $+20m\,\text{dB/dec} \Rightarrow s^m$ in the numerator; pair each s with a lowest pole as $\frac{s}{s+p}$.
3. *Assemble* $H(s) = K \cdot \prod(\text{factors})$ with $\omega_c = 2\pi f_c$.
4. *Find K:* at the marked frequency, add the straight-line dB of every factor; $20\log_{10} K =$ marked level $-$ that sum.
5. *Check:* the sum of the factor plots must reproduce the given plot.
6. *Output:* evaluate the exact factors at $s = j\omega_{in}$, multiply magnitudes, add angles, apply T5.

Problems

Easy problems

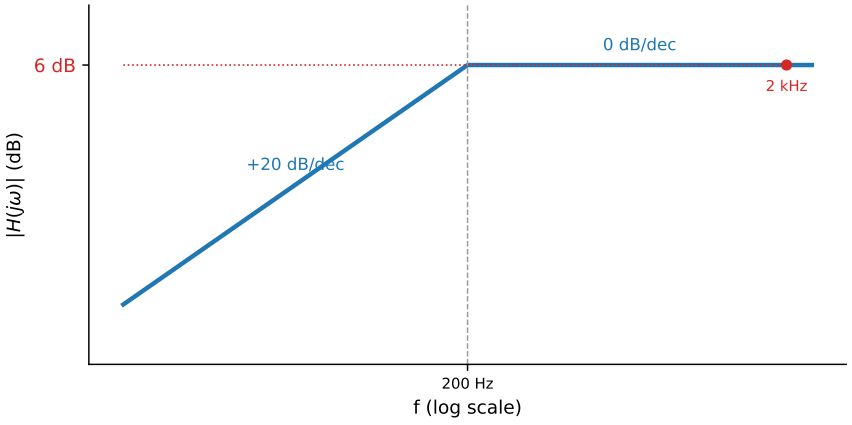
Problem 1D.1 — Easy

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.5 \sin(2\pi \cdot 2000\, t)$.



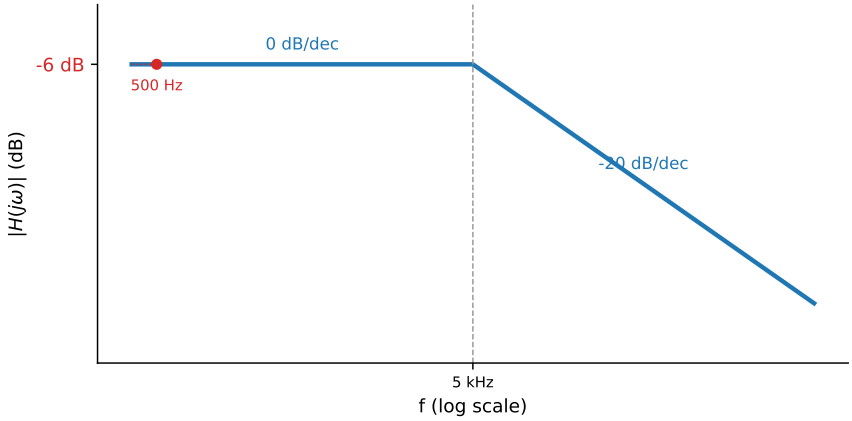
Problem 1D.2 — Easy

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 1 \sin(2\pi \cdot 100\, t)$.



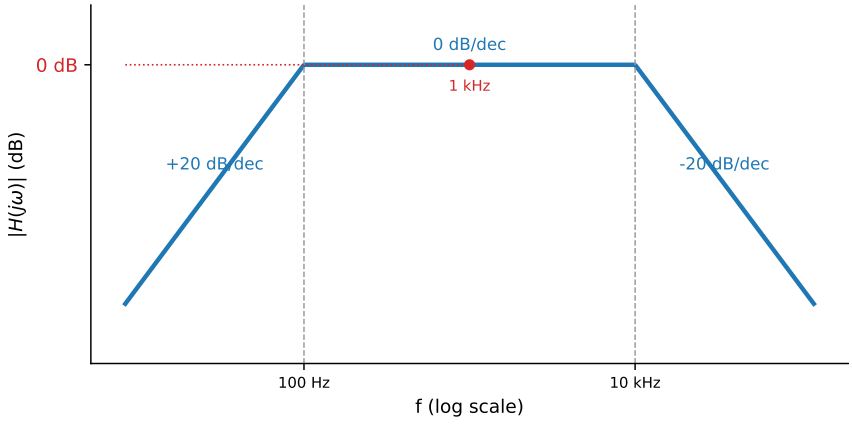
Problem 1D.3 — Easy

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 2\cos(2\pi \cdot 10000\, t)$.



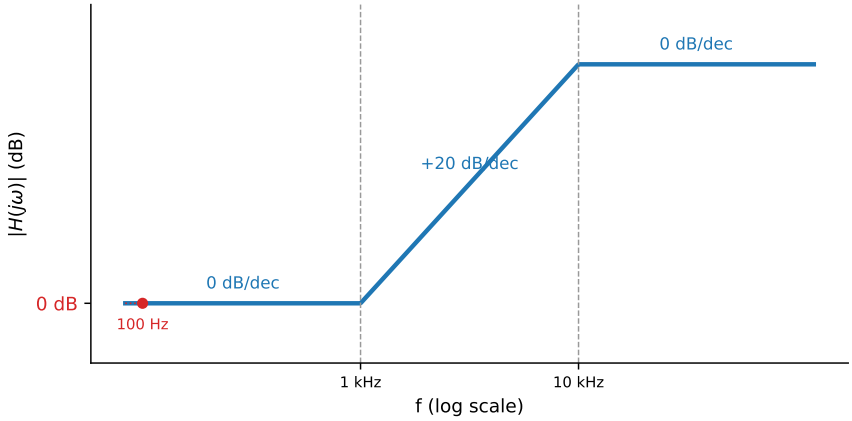
Problem 1D.4 — Easy

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\, t)$.



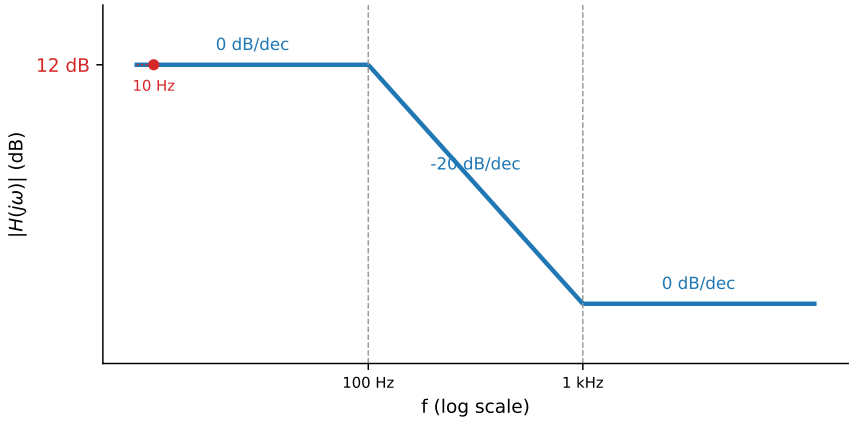
Problem 1D.5 — Easy

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 3000\,t)$.



Problem 1D.6 — Easy

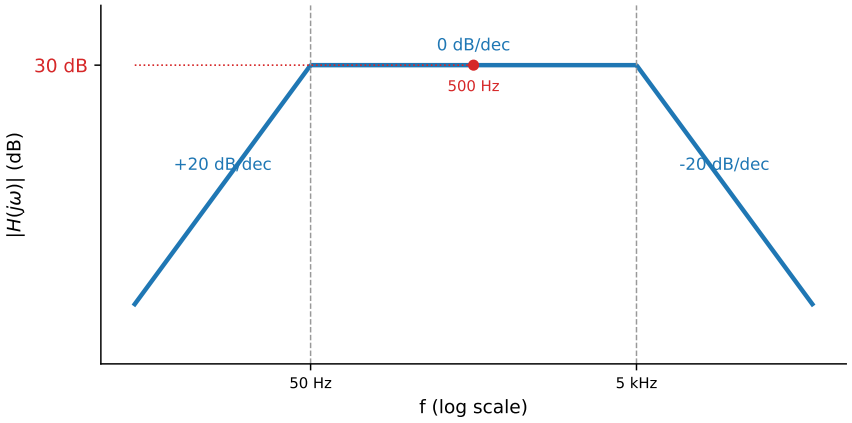
The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 500\,t)$.



Medium problems

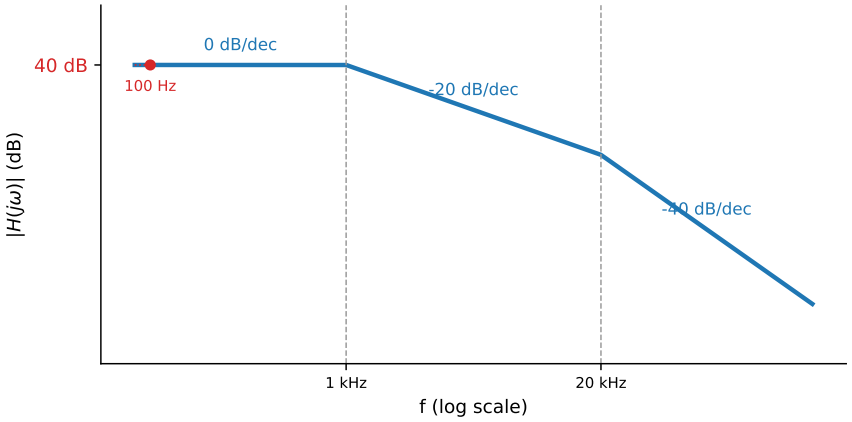
Problem 1D.7 — Medium

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 20000\,t)$.



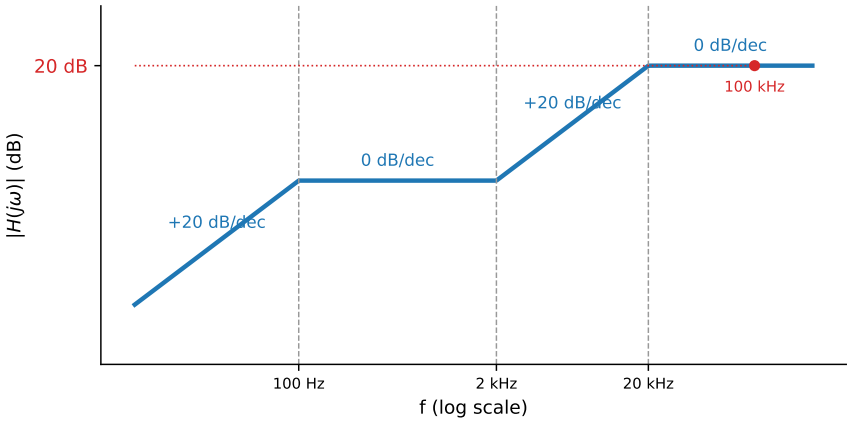
Problem 1D.8 — Medium

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 5000\,t)$.



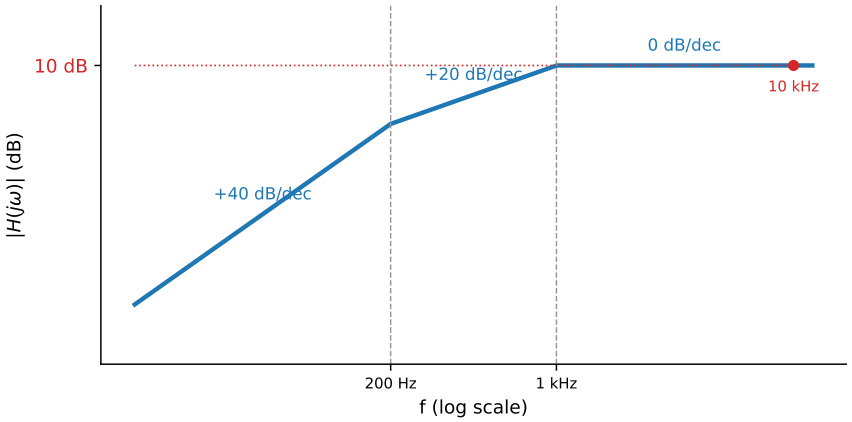
Problem 1D.9 — Medium

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 1000\,t)$.



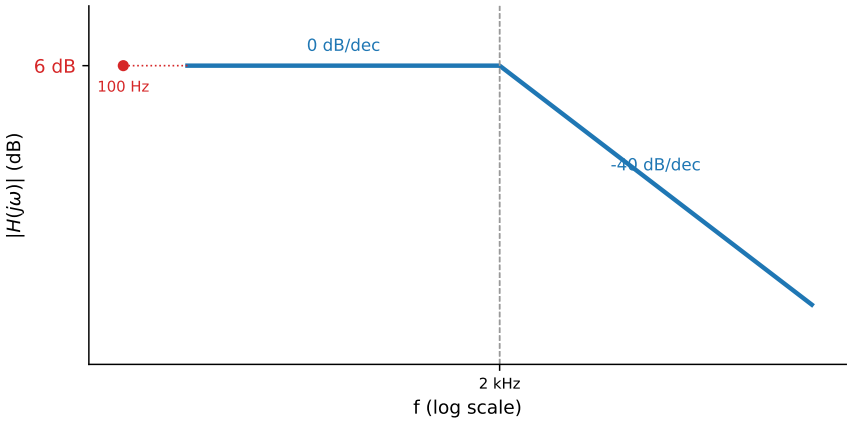
Problem 1D.10 — Medium

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 400\,t)$.



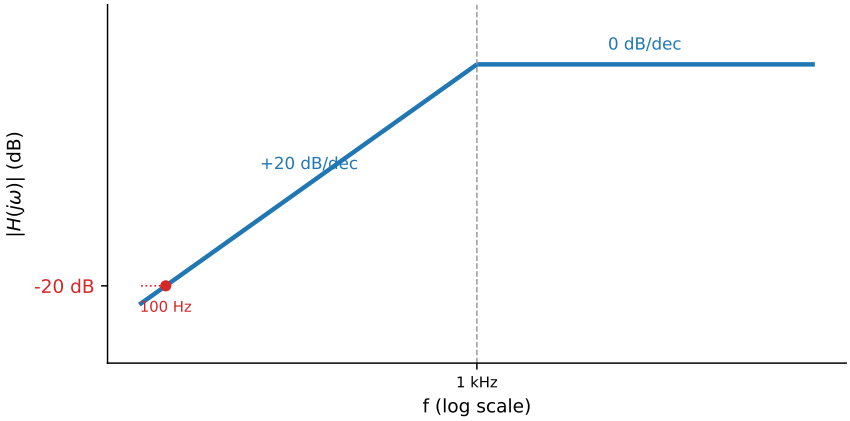
Problem 1D.11 — Medium

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 1 \sin(2\pi \cdot 4000\, t)$.



Problem 1D.12 — Medium

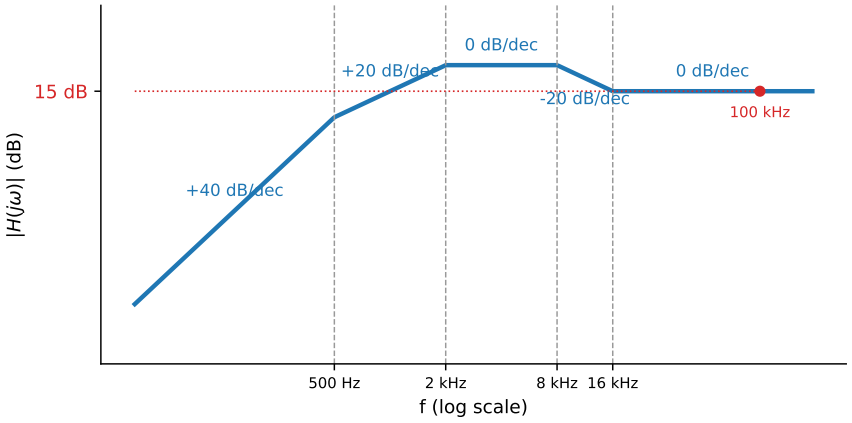
The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 1 \sin\left(2\pi \cdot 2000\, t + \frac{\pi}{4}\right)$.



Hard problems

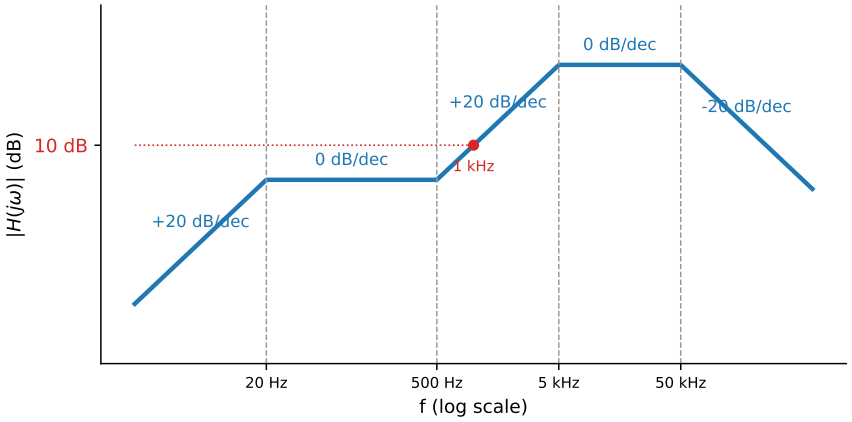
Problem 1D.13 — Hard

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.3\sin(2\pi \cdot 10000\,t)$.



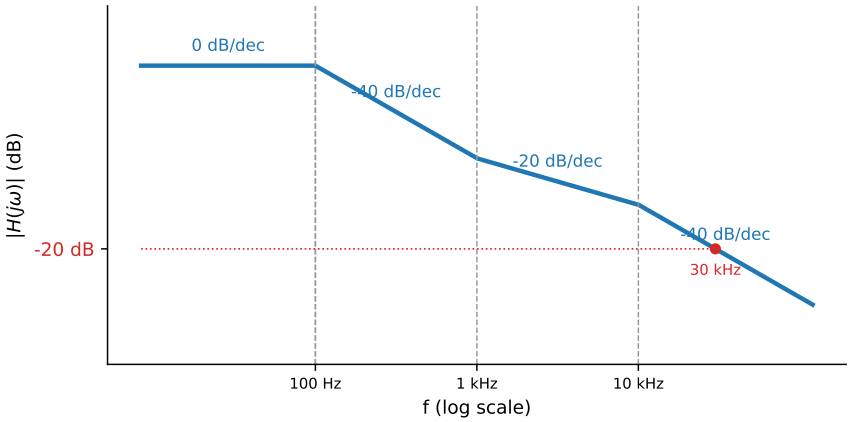
Problem 1D.14 — Hard

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin\left(2\pi \cdot 2000\,t + \frac{\pi}{3}\right)$.



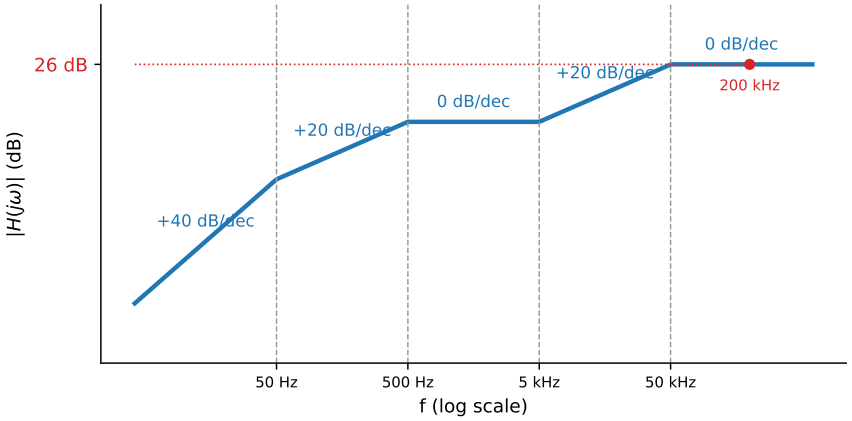
Problem 1D.15 — Hard

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin(2\pi \cdot 300\, t)$.



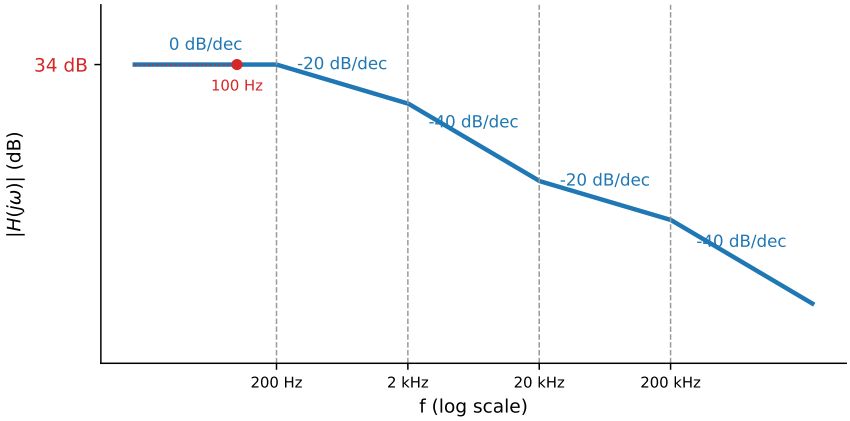
Problem 1D.16 — Hard

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \cos\left(2\pi \cdot 1000\, t - \frac{\pi}{6}\right)$.



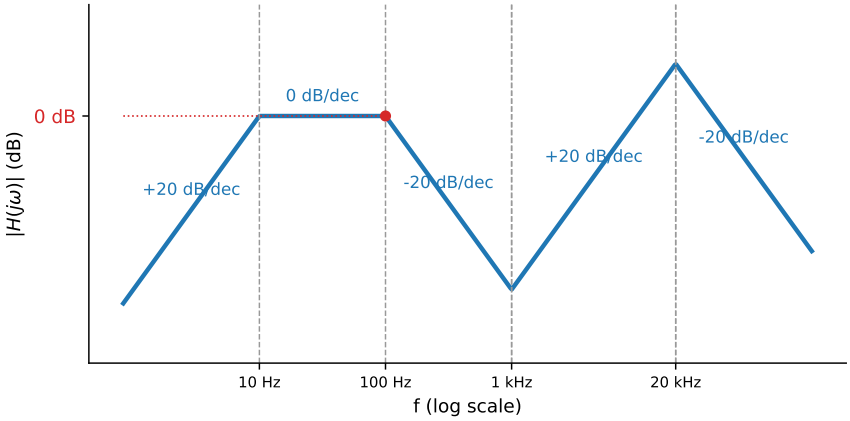
Problem 1D.17 — Hard

The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.02\sin\left(2\pi \cdot 5000\,t + \frac{\pi}{2}\right)$.



Problem 1D.18 — Hard

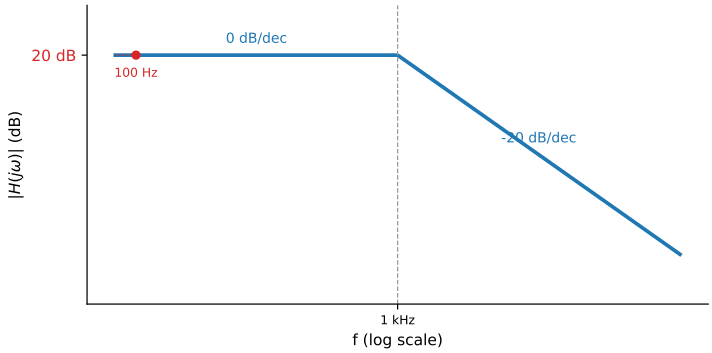
The straight-line (asymptotic) magnitude Bode plot of $H(s) = \dfrac{V_o(s)}{V_i(s)}$ is shown. (a) Find $H(s)$ (assume $K > 0$ and no right-half-plane poles or zeros). (b) Find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin\left(2\pi \cdot 3000\,t\right)$.



Solutions

Every solution follows the same six steps of the pipeline card.

Solution 1D.1 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
1 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$	1 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

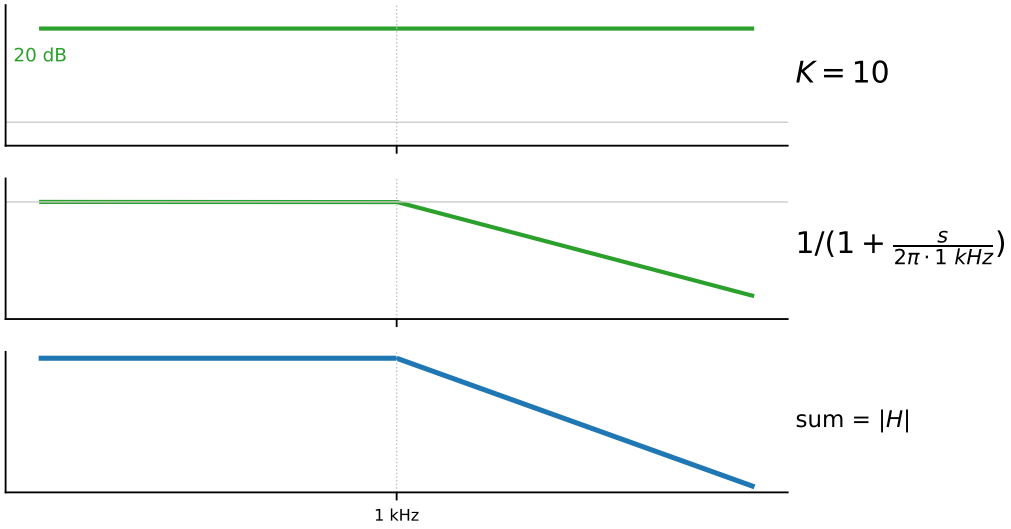
$$20\log_{10} K = 20 - (0) = 20\ \text{dB} \quad \Rightarrow \quad K = 10^{20/20} = 10$$

So

$$\boxed{H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 1$ kHz	0 dB/dec	20 dB at end
$f > 1$ kHz	−20 dB/dec	20 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi\cdot 2000\,t)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$	0.4472	−1.11
total	**4.472** (product)	**−1.11** (sum)

$$|H(j\omega_{in})| = 4.47, \qquad \angle H(j\omega_{in}) = -1.11 \text{ rad}$$

$$v_{o,ac}(t) = 4.47 \times 0.5 \sin\left(2\pi \cdot 2000\, t - 1.11\right) = 2.24 \sin\left(2\pi \cdot 2000\, t - 1.11\right)$$

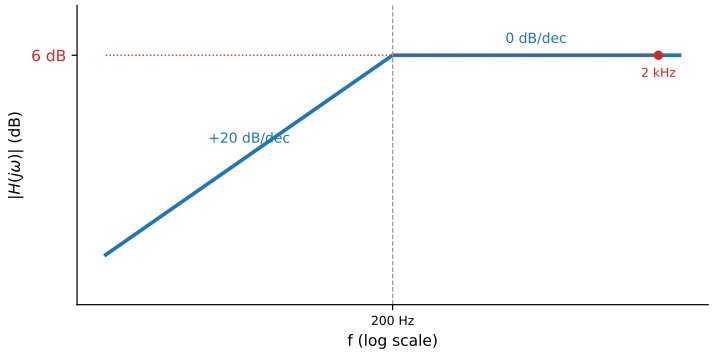
$$\boxed{V_o(t) = 2.24 \sin\left(2\pi \cdot 2000\, t - 1.11\right) \text{ V}}$$

Answer

$$H(s) = 10 \frac{1}{1 + \dfrac{s}{2\pi \cdot 1\text{k}}} \quad \text{(corners: 1 kHz)}$$

$$V_o(t) = 2.24 \sin\left(2\pi \cdot 2000\, t - 1.11\right) \text{ V}$$

Solution 1D.2 — Easy



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
200 Hz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (200 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s + 2\pi\cdot 200}$$

Step 4 — Find K from the marked level

The plot marks the level 6 dB at $f = 2$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 2$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi\cdot 200}$	200 Hz	$f > f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

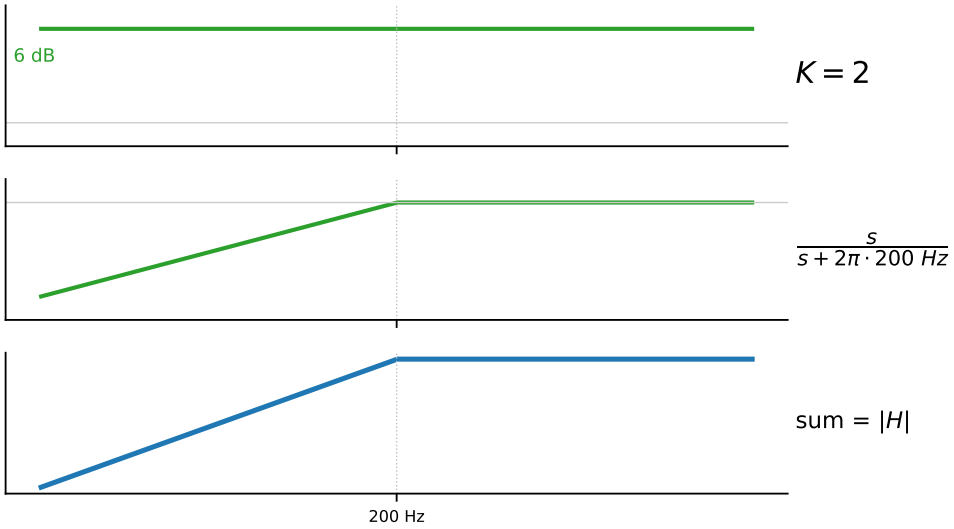
$$20\log_{10} K = 6 - (0) = 6 \text{ dB} \quad \Rightarrow \quad K = 10^{6/20} = 2$$

So

$$\boxed{H(s) = 2\,\frac{s}{s + 2\pi \cdot 200}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 200$ Hz	+20 dB/dec	6 dB at end
$f > 200$ Hz	0 dB/dec	6 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 100\,t)$, so $\omega_{in} = 2\pi \cdot 100 = 628\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
2	1.995	0
$\dfrac{s}{s + 2\pi \cdot 200}$	0.4472	1.11
total	**0.8923** (product)	**1.11** (sum)

$$|H(j\omega_{in})| = 0.892, \qquad \angle H(j\omega_{in}) = 1.11 \text{ rad}$$
$$v_{o,ac}(t) = 0.892 \times 1 \sin(2\pi \cdot 100\,t + 1.11) = 0.892 \sin(2\pi \cdot 100\,t + 1.11)$$

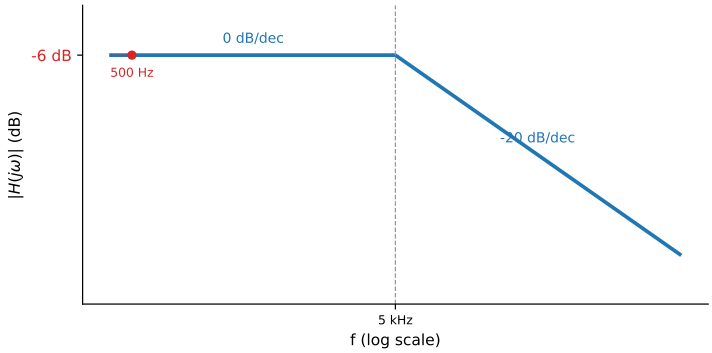
$$\boxed{V_o(t) = 0.892 \sin\left(2\pi \cdot 100\, t + 1.11\right) \text{ V}}$$

Answer

$$H(s) = 2\,\frac{s}{s + 2\pi \cdot 200} \quad \text{(corners: 200 Hz)}$$

$$V_o(t) = 0.892 \sin\left(2\pi \cdot 100\, t + 1.11\right) \text{ V}$$

Solution 1D.3 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
5 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level -6 dB at $f = 500$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 500$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$	5 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

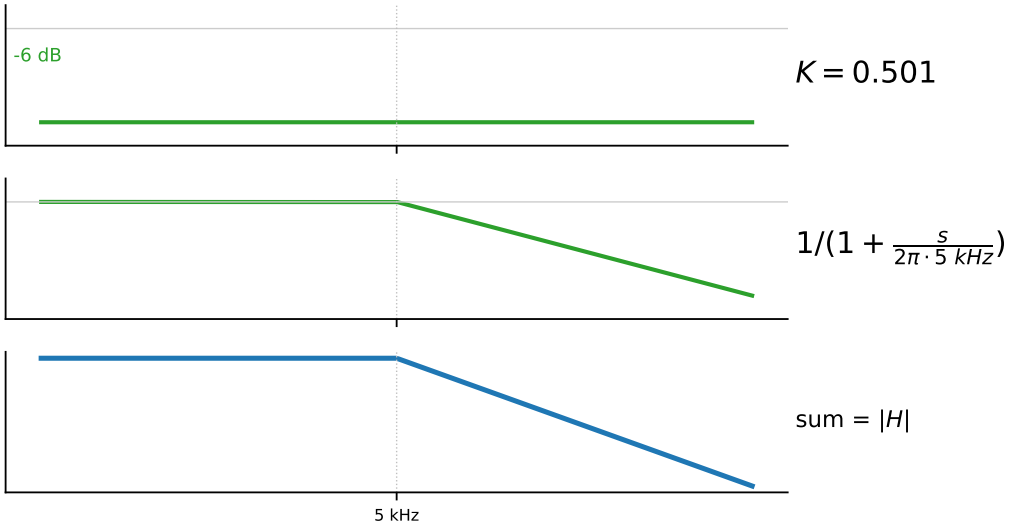
$$20\log_{10} K = -6 - (0) = -6 \text{ dB} \quad \Rightarrow \quad K = 10^{-6/20} = 0.501$$

So

$$\boxed{H(s) = 0.501 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 5$ kHz	0 dB/dec	-6 dB at end
$f > 5$ kHz	-20 dB/dec	-6 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 2\cos(2\pi \cdot 10000\, t)$, so $\omega_{in} = 2\pi \cdot 10000 = 62800$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.501	0.5012	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}$	0.4472	-1.11
total	**0.2241** (product)	$\mathbf{-1.11}$ (sum)

$$|H(j\omega_{in})| = 0.224, \qquad \angle H(j\omega_{in}) = -1.11 \text{ rad}$$

$$v_{o,ac}(t) = 0.224 \times 2 \cos\left(2\pi \cdot 10000\, t - 1.11\right) = 0.448 \cos\left(2\pi \cdot 10000\, t - 1.11\right)$$

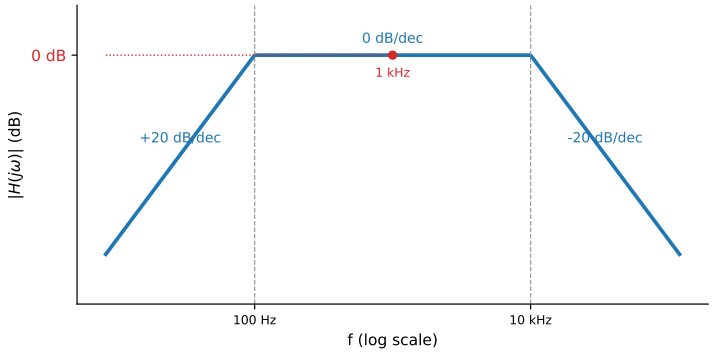
$$\boxed{V_o(t) = 0.448 \cos\left(2\pi \cdot 10000\, t - 1.11\right) \text{ V}}$$

Answer

$$H(s) = 0.501 \frac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}} \quad \text{(corners: 5 kHz)}$$

$$V_o(t) = 0.448 \cos\left(2\pi \cdot 10000\, t - 1.11\right) \text{ V}$$

Solution 1D.4 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	$+20$	0	-20	one pole at f_c
10 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (100 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 100}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 0 dB at $f = 1$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 1$ kHz	asymptote (dB)
$\dfrac{s}{s+2\pi\cdot 100}$	100 Hz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	10 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

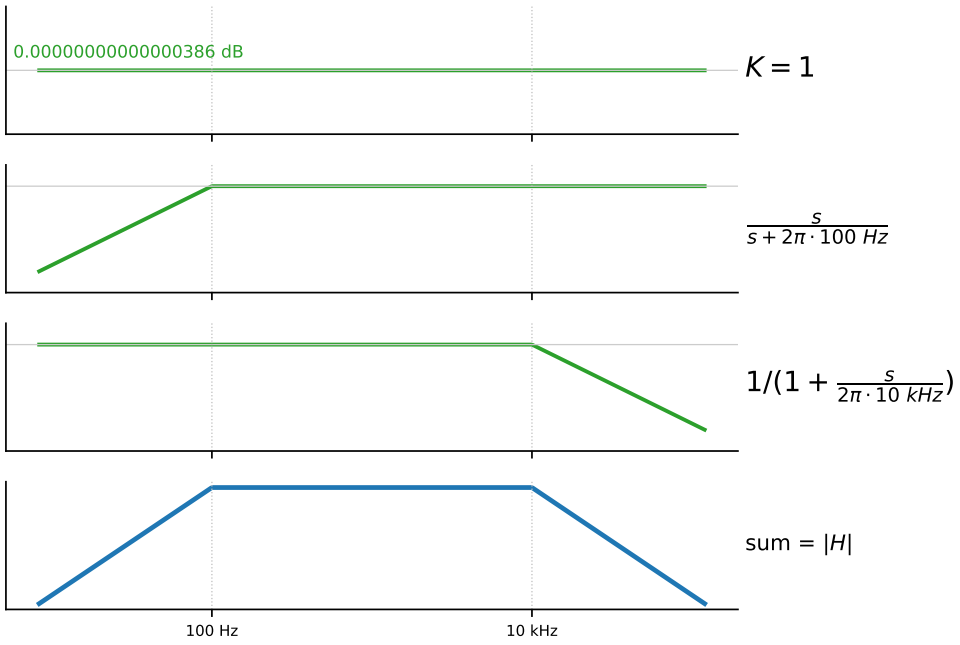
$$20\log_{10} K = 0 - (0) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1\,\frac{s}{s + 2\pi\cdot 100}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 100$ Hz	+20 dB/dec	0 dB at end
100 Hz – 10 kHz	0 dB/dec	0 dB at start, 0 dB at end
$f > 10$ kHz	−20 dB/dec	0 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 100}$	0.995	0.0997
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	0.995	-0.0997
total	**0.9901** (product)	**0** (sum)

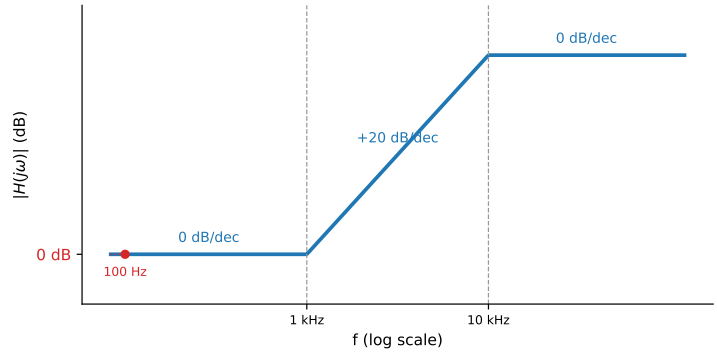
$$|H(j\omega_{in})| = 0.99, \qquad \angle H(j\omega_{in}) = 0 \text{ rad}$$
$$v_{o,ac}(t) = 0.99 \times 1 \sin(2\pi\cdot 1000\,t + 0) = 0.99 \sin(2\pi\cdot 1000\,t + 0)$$

$$\boxed{V_o(t) = 0.99 \sin(2\pi\cdot 1000\,t + 0) \text{ V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 100}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}} \quad \text{(corners: 100 Hz, 10 kHz)}$$
$$V_o(t) = 0.99 \sin(2\pi\cdot 1000\,t + 0) \text{ V}$$

Solution 1D.5 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
1 kHz	0	+20	+20	one zero at f_c
10 kHz	+20	0	−20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 0 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	10 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

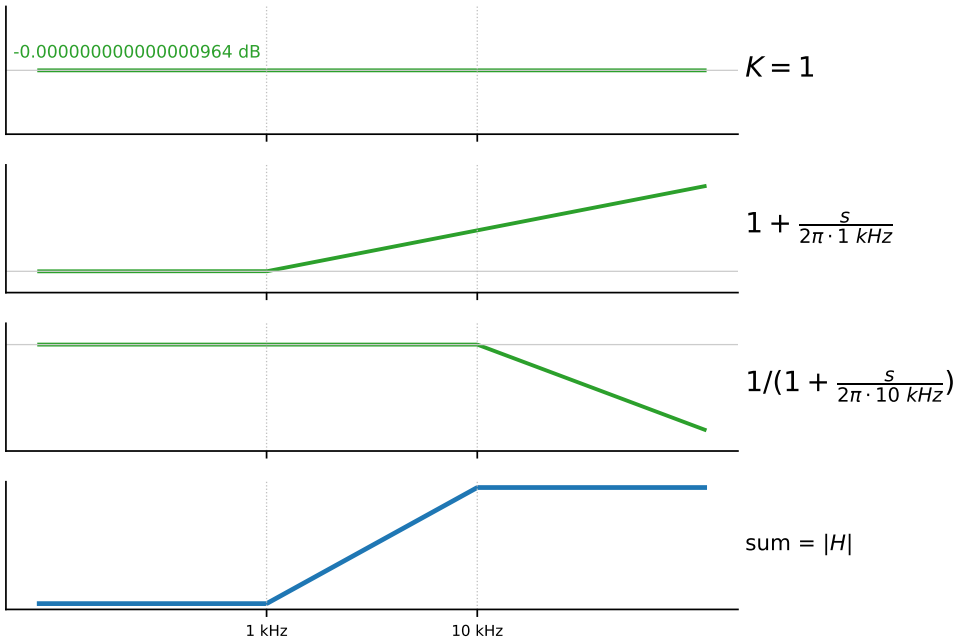
$$20\log_{10} K = 0 - (0) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1 \left(1 + \frac{s}{2\pi \cdot 1\text{k}}\right) \frac{1}{1 + \dfrac{s}{2\pi \cdot 10\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 1$ kHz	0 dB/dec	0 dB at end
1 kHz – 10 kHz	+20 dB/dec	0 dB at start, 20 dB at end
$f > 10$ kHz	0 dB/dec	20 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 3000\, t)$, so $\omega_{in} = 2\pi \cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	3.162	1.25
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	0.9578	-0.291
total	**3.029** (product)	**0.958** (sum)

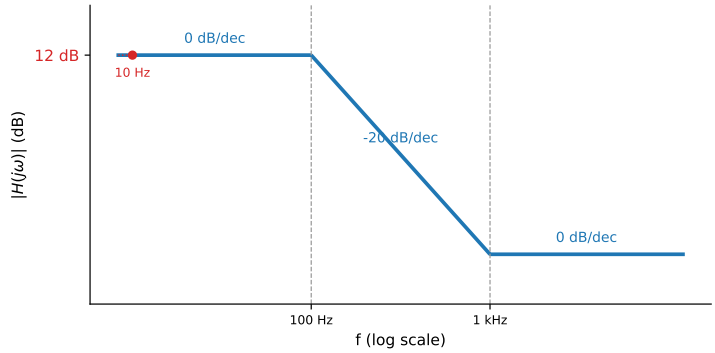
$$|H(j\omega_{in})| = 3.03, \qquad \angle H(j\omega_{in}) = 0.958 \text{ rad}$$
$$v_{o,ac}(t) = 3.03 \times 0.5 \sin\left(2\pi\cdot 3000\,t + 0.958\right) = 1.51 \sin\left(2\pi\cdot 3000\,t + 0.958\right)$$

$$\boxed{V_o(t) = 1.51 \sin\left(2\pi\cdot 3000\,t + 0.958\right)\ \text{V}}$$

Answer

$$H(s) = 1\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}} \quad \text{(corners: 1 kHz, 10 kHz)}$$
$$V_o(t) = 1.51 \sin\left(2\pi\cdot 3000\,t + 0.958\right)\ \text{V}$$

Solution 1D.6 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	0	-20	-20	one pole at f_c
1 kHz	-20	0	$+20$	one zero at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)$$

Step 4 — Find K from the marked level

The plot marks the level 12 dB at $f = 10$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 10$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	100 Hz	$f < f_c$: ≈ 1	0
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

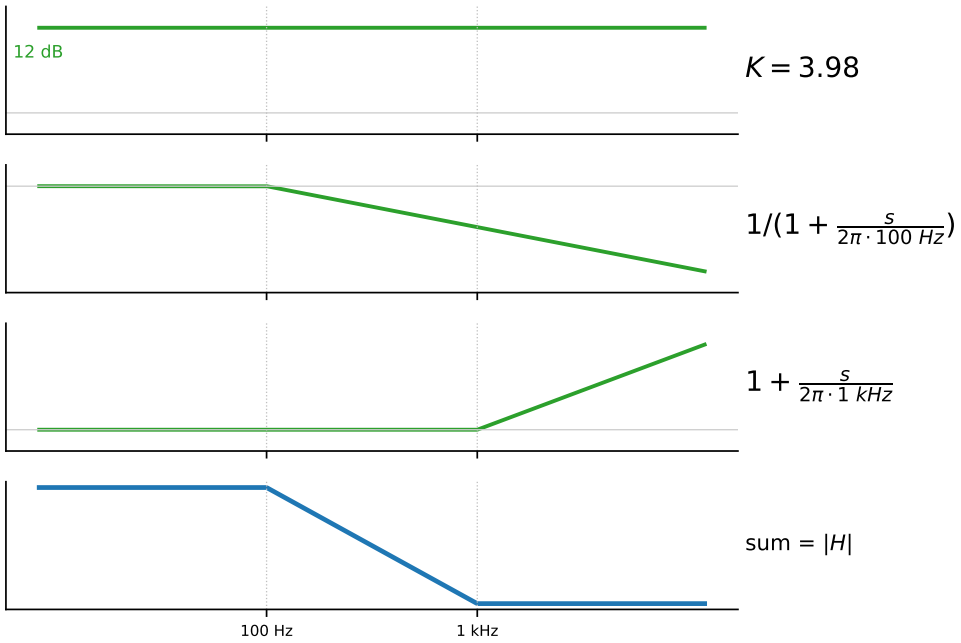
$$20\log_{10} K = 12 - (0) = 12 \text{ dB} \quad \Rightarrow \quad K = 10^{12/20} = 3.98$$

So

$$\boxed{H(s) = 3.98\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 100$ Hz	0 dB/dec	12 dB at end
100 Hz – 1 kHz	−20 dB/dec	12 dB at start, -8 dB at end
$f > 1$ kHz	0 dB/dec	-8 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 500\,t)$, so $\omega_{in} = 2\pi\cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
3.98	3.981	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	0.1961	-1.37
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1.118	0.464
total	**0.8729** (product)	$-$**0.91** (sum)

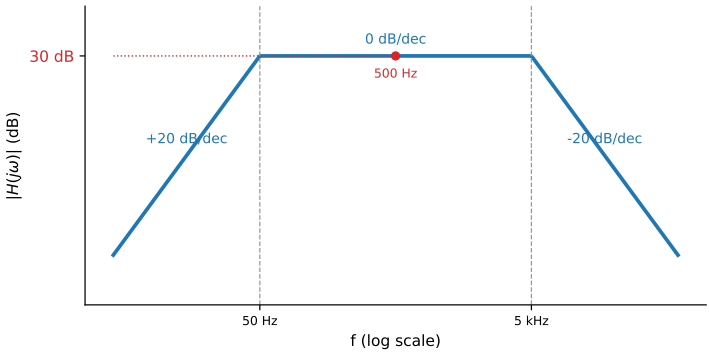
$$|H(j\omega_{in})| = 0.873, \qquad \angle H(j\omega_{in}) = -0.91 \text{ rad}$$
$$v_{o,ac}(t) = 0.873 \times 1 \sin\left(2\pi\cdot 500\,t - 0.91\right) = 0.873 \sin\left(2\pi\cdot 500\,t - 0.91\right)$$

$$\boxed{V_o(t) = 0.873 \sin\left(2\pi\cdot 500\,t - 0.91\right)\ \text{V}}$$

Answer

$$H(s) = 3.98\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right) \quad \text{(corners: 100 Hz, 1 kHz)}$$
$$V_o(t) = 0.873 \sin\left(2\pi\cdot 500\,t - 0.91\right)\ \text{V}$$

Solution 1D.7 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
50 Hz	$+20$	0	-20	one pole at f_c
5 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (50 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 50}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 30 dB at $f = 500$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 500$ Hz	asymptote (dB)
$\dfrac{s}{s+2\pi\cdot 50}$	50 Hz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$	5 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

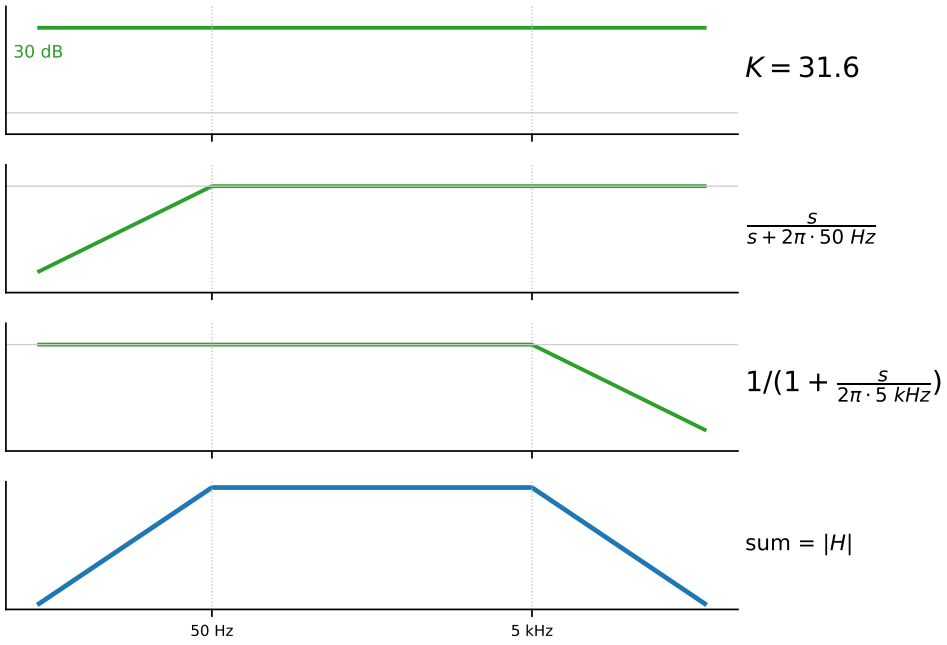
$$20\log_{10} K = 30 - (0) = 30 \text{ dB} \quad \Rightarrow \quad K = 10^{30/20} = 31.6$$

So

$$\boxed{H(s) = 31.6\,\frac{s}{s + 2\pi\cdot 50}\,\frac{1}{1 + \dfrac{s}{2\pi\cdot 5\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 50$ Hz	+20 dB/dec	30 dB at end
50 Hz – 5 kHz	0 dB/dec	30 dB at start, 30 dB at end
$f > 5$ kHz	−20 dB/dec	30 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi\cdot 20000\,t)$, so $\omega_{in} = 2\pi\cdot 20000 = 1.26\times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
31.6	31.62	0
$\dfrac{s}{s + 2\pi \cdot 50}$	1	0.0025
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}$	0.2425	-1.33
total	**7.67** (product)	$-$**1.32** (sum)

$$|H(j\omega_{in})| = 7.67, \qquad \angle H(j\omega_{in}) = -1.32 \text{ rad}$$
$$v_{o,ac}(t) = 7.67 \times 0.1 \sin(2\pi \cdot 20000\, t - 1.32) = 0.767 \sin(2\pi \cdot 20000\, t - 1.32)$$

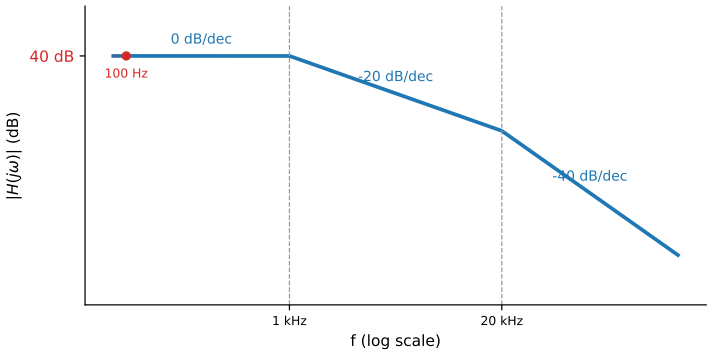
$$\boxed{V_o(t) = 0.767 \sin(2\pi \cdot 20000\, t - 1.32) \text{ V}}$$

Answer

$H(s) = 31.6 \dfrac{s}{s + 2\pi \cdot 50} \dfrac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}$ (corners: 50 Hz, 5 kHz)

$V_o(t) = 0.767 \sin(2\pi \cdot 20000\, t - 1.32)$ V

Solution 1D.8 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
1 kHz	0	-20	-20	one pole at f_c
20 kHz	-20	-40	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 40 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	20 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

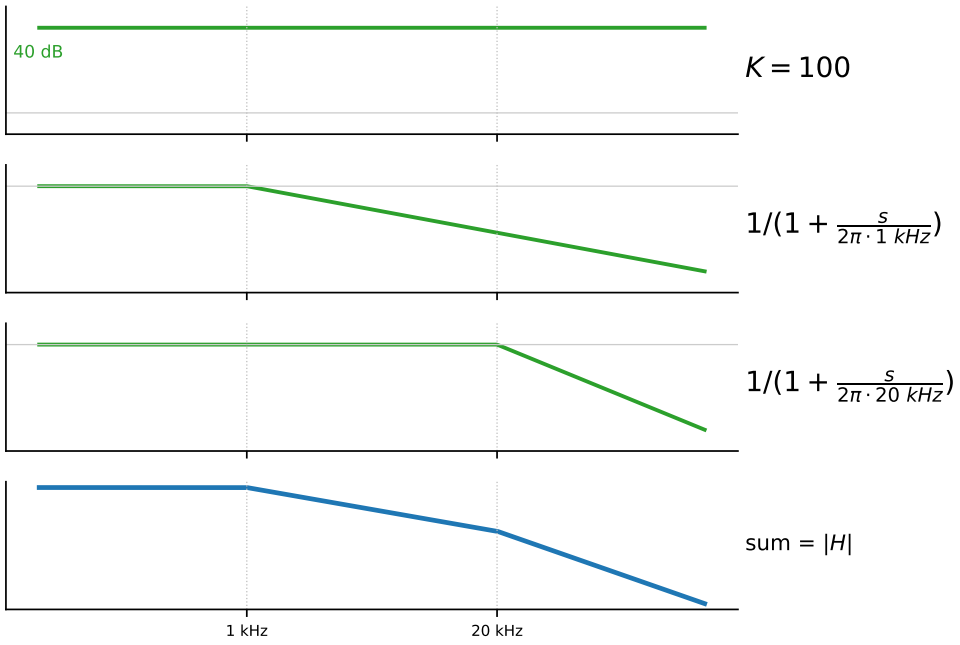
$$20\log_{10} K = 40 - (0) = 40 \text{ dB} \quad \Rightarrow \quad K = 10^{40/20} = 100$$

So

$$\boxed{H(s) = 100\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 1$ kHz	0 dB/dec	40 dB at end
1 kHz – 20 kHz	−20 dB/dec	40 dB at start, 14 dB at end
$f > 20$ kHz	−40 dB/dec	14 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi\cdot 5000\,t)$, so $\omega_{in} = 2\pi\cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
100	100	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$	0.1961	-1.37
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9701	-0.245
total	**19.03** (product)	$-$**1.62** (sum)

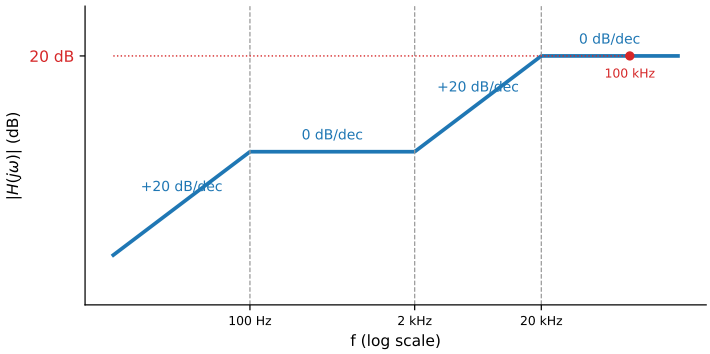
$$|H(j\omega_{in})| = 19, \qquad \angle H(j\omega_{in}) = -1.62 \text{ rad}$$
$$v_{o,ac}(t) = 19 \times 0.05 \sin(2\pi\cdot 5000\,t - 1.62) = 0.951 \sin(2\pi\cdot 5000\,t - 1.62)$$

$$\boxed{V_o(t) = 0.951 \sin(2\pi\cdot 5000\,t - 1.62) \text{ V}}$$

Answer

$$H(s) = 100\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}} \quad \text{(corners: 1 kHz, 20 kHz)}$$
$$V_o(t) = 0.951 \sin(2\pi\cdot 5000\,t - 1.62) \text{ V}$$

Solution 1D.9 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	$+20$	0	-20	one pole at f_c
2 kHz	0	$+20$	$+20$	one zero at f_c
20 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (100 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 100}\left(1+\frac{s}{2\pi\cdot 2\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 100$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 100}$	100 Hz	$f > f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 2\text{k}}\right)$	2 kHz	$f > f_c$: $\approx \frac{\omega}{z}$	34
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}$	20 kHz	$f > f_c$: $\approx \frac{p}{\omega}$	-14
sum			**20**

The dB values of the factors *add* (magnitudes multiply), so

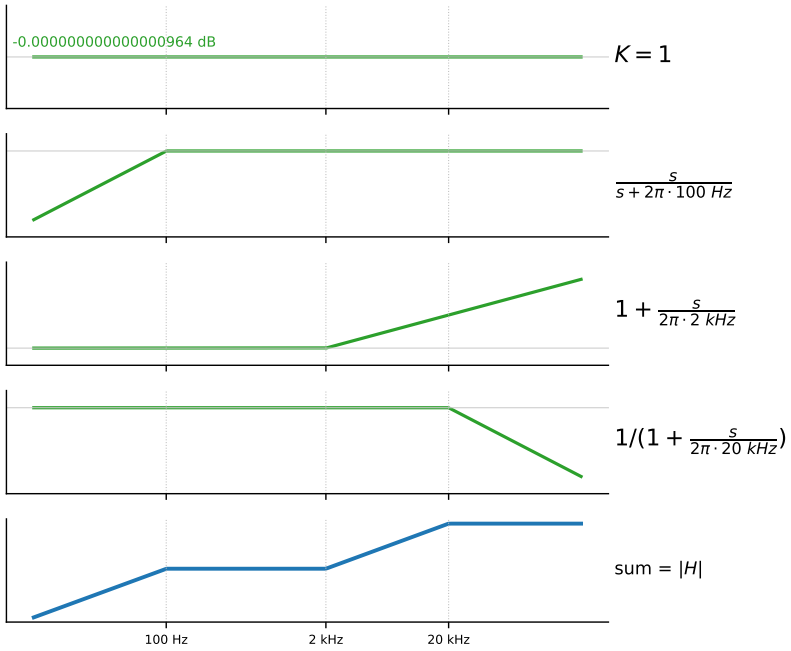
$$20 \log_{10} K = 20 - (20) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1 \, \frac{s}{s + 2\pi \cdot 100} \left(1 + \frac{s}{2\pi \cdot 2\text{k}}\right) \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 100$ Hz	+20 dB/dec	0 dB at end
100 Hz – 2 kHz	0 dB/dec	0 dB at start, 0 dB at end
2 kHz – 20 kHz	+20 dB/dec	0 dB at start, 20 dB at end
$f > 20$ kHz	0 dB/dec	20 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 100}$	0.995	0.0997
$\left(1+\dfrac{s}{2\pi\cdot 2\text{k}}\right)$	1.118	0.464
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9988	−0.05
total	**1.111** (product)	**0.513** (sum)

$$|H(j\omega_{in})| = 1.11, \qquad \angle H(j\omega_{in}) = 0.513 \text{ rad}$$
$$v_{o,ac}(t) = 1.11 \times 0.2\,\sin(2\pi\cdot 1000\,t + 0.513) = 0.222\,\sin(2\pi\cdot 1000\,t + 0.513)$$

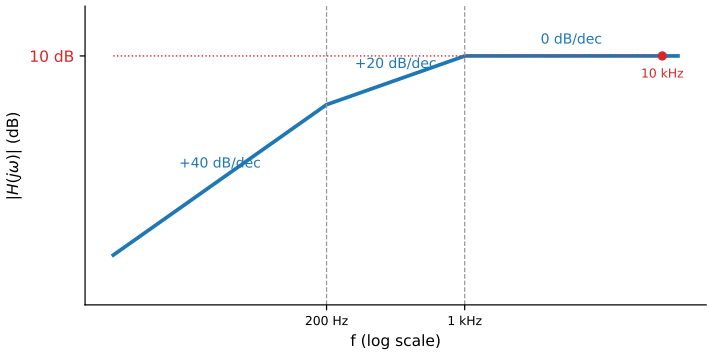
$$\boxed{V_o(t) = 0.222\,\sin(2\pi\cdot 1000\,t + 0.513)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 100}\left(1+\frac{s}{2\pi\cdot 2\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}} \quad \text{(corners: 100 Hz, 2 kHz, 20 kHz)}$$

$$V_o(t) = 0.222\,\sin(2\pi\cdot 1000\,t + 0.513)\ \text{V}$$

Solution 1D.10 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
200 Hz	$+40$	$+20$	-20	one pole at f_c
1 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+40$ dB/dec, i.e. $|H| \propto \omega^2$ at low frequency: the numerator contains s^2. As in class, pair each s with one of the *lowest* poles (200 Hz, 1 kHz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 200}\,\frac{s}{s+2\pi\cdot 1\text{k}}$$

Step 4 — Find K from the marked level

The plot marks the level 10 dB at $f = 10$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 10$ kHz	asymptote (dB)
$\frac{s}{s+2\pi\cdot 200}$	200 Hz	$f > f_c$: ≈ 1	0
$\frac{s}{s+2\pi\cdot 1\text{k}}$	1 kHz	$f > f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

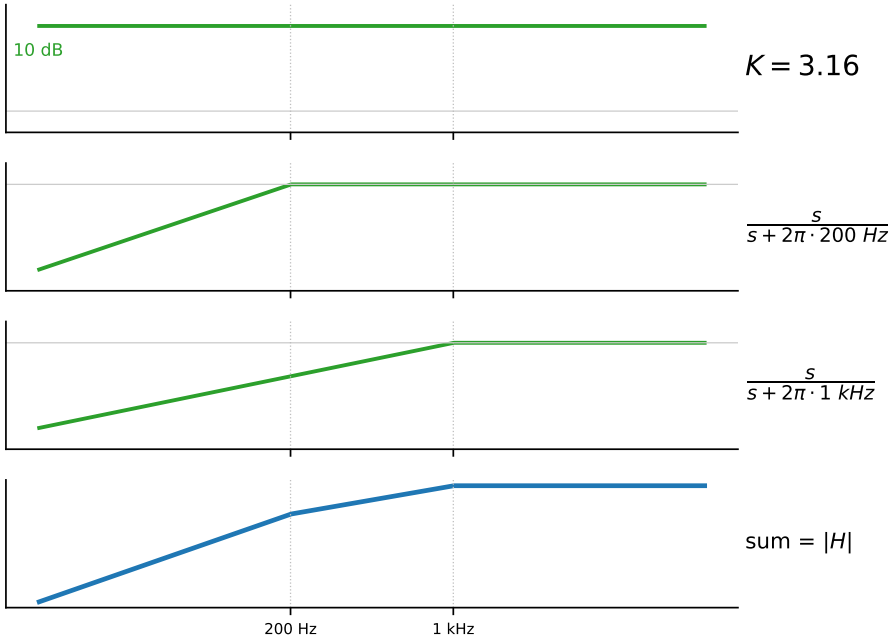
$$20\log_{10} K = 10 - (0) = 10 \text{ dB} \quad \Rightarrow \quad K = 10^{10/20} = 3.16$$

So

$$\boxed{H(s) = 3.16\,\frac{s}{s + 2\pi\cdot 200}\,\frac{s}{s + 2\pi\cdot 1\text{k}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 200$ Hz	+40 dB/dec	-3.98 dB at end
200 Hz – 1 kHz	+20 dB/dec	-3.98 dB at start, 10 dB at end
$f > 1$ kHz	0 dB/dec	10 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi\cdot 400\,t)$, so $\omega_{in} = 2\pi\cdot 400 = 2510\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
3.16	3.162	0
$\dfrac{s}{s+2\pi\cdot 200}$	0.8944	0.464
$\dfrac{s}{s+2\pi\cdot 1\text{k}}$	0.3714	1.19
total	**1.05** (product)	**1.65** (sum)

$$|H(j\omega_{in})| = 1.05, \qquad \angle H(j\omega_{in}) = 1.65 \text{ rad}$$
$$v_{o,ac}(t) = 1.05 \times 0.5 \sin(2\pi\cdot 400\,t + 1.65) = 0.525 \sin(2\pi\cdot 400\,t + 1.65)$$

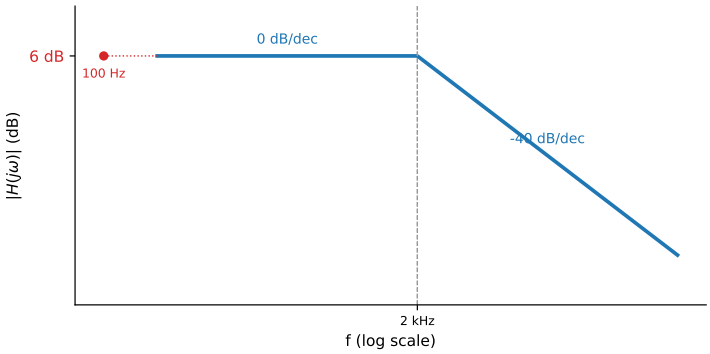
$$\boxed{V_o(t) = 0.525 \sin(2\pi\cdot 400\,t + 1.65) \text{ V}}$$

Answer

$H(s) = 3.16 \dfrac{s}{s+2\pi\cdot 200}\dfrac{s}{s+2\pi\cdot 1\text{k}}$ (corners: 200 Hz, 1 kHz)

$V_o(t) = 0.525 \sin(2\pi\cdot 400\,t + 1.65)$ V

Solution 1D.11 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
2 kHz	0	-40	-40	two poles at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 6 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$	2 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$	2 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

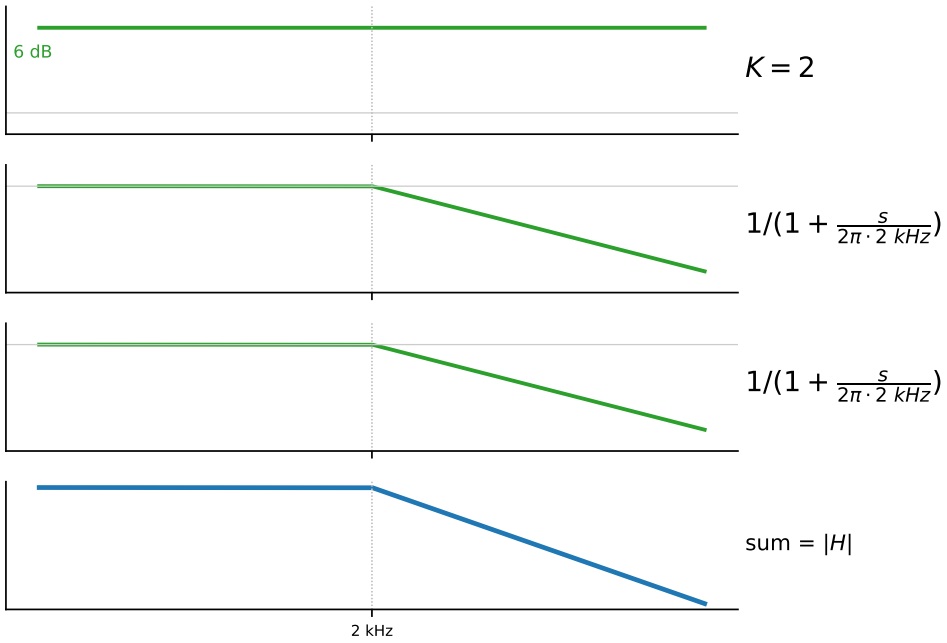
$$20\log_{10} K = 6 - (0) = 6\ \text{dB} \quad \Rightarrow \quad K = 10^{6/20} = 2$$

So

$$\boxed{H(s) = 2\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 2$ kHz	0 dB/dec	6 dB at end
$f > 2$ kHz	-40 dB/dec	6 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 4000\,t)$, so $\omega_{in} = 2\pi\cdot 4000 = 25100\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
2	1.995	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$	0.4472	-1.11
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$	0.4472	-1.11
total	**0.3991** (product)	$-$**2.21** (sum)

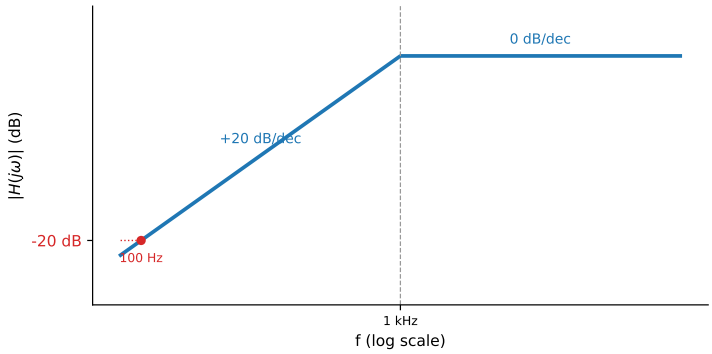
$$|H(j\omega_{in})| = 0.399, \qquad \angle H(j\omega_{in}) = -2.21 \text{ rad}$$
$$v_{o,ac}(t) = 0.399 \times 1 \sin\left(2\pi\cdot 4000\,t - 2.21\right) = 0.399 \sin\left(2\pi\cdot 4000\,t - 2.21\right)$$

$$\boxed{V_o(t) = 0.399 \sin\left(2\pi\cdot 4000\,t - 2.21\right) \text{ V}}$$

Answer

$$H(s) = 2\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}} \quad \text{(corners: 2 kHz, 2 kHz)}$$
$$V_o(t) = 0.399 \sin\left(2\pi\cdot 4000\,t - 2.21\right) \text{ V}$$

Solution 1D.12 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
1 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (1 kHz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s + 2\pi\cdot 1\text{k}}$$

Step 4 — Find K from the marked level

The plot marks the level -20 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\frac{s}{s+2\pi\cdot 1\text{k}}$	1 kHz	$f < f_c$: $\approx \frac{\omega}{p}$	-20
sum			$\mathbf{-20}$

The dB values of the factors *add* (magnitudes multiply), so

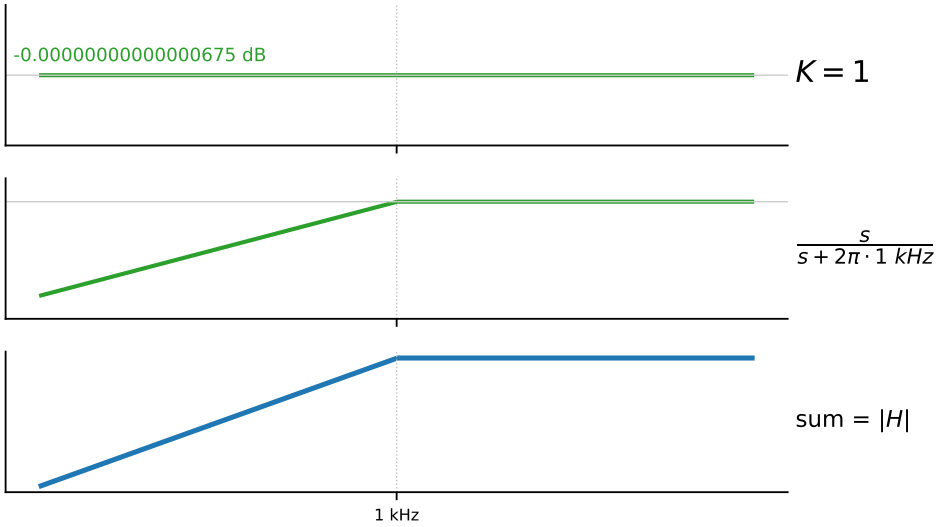
$$20\log_{10} K = -20 - (-20) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1\,\frac{s}{s + 2\pi \cdot 1\text{k}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 1$ kHz	+20 dB/dec	0 dB at end
$f > 1$ kHz	0 dB/dec	0 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 1 \sin\left(2\pi \cdot 2000\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s + 2\pi \cdot 1\text{k}}$	0.8944	0.464
total	**0.8944** (product)	**0.464** (sum)

$$|H(j\omega_{in})| = 0.894, \qquad \angle H(j\omega_{in}) = 0.464 \text{ rad}$$
$$v_{o,ac}(t) = 0.894 \times 1 \sin\left(2\pi \cdot 2000\,t + 1.25\right) = 0.894 \sin\left(2\pi \cdot 2000\,t + 1.25\right)$$

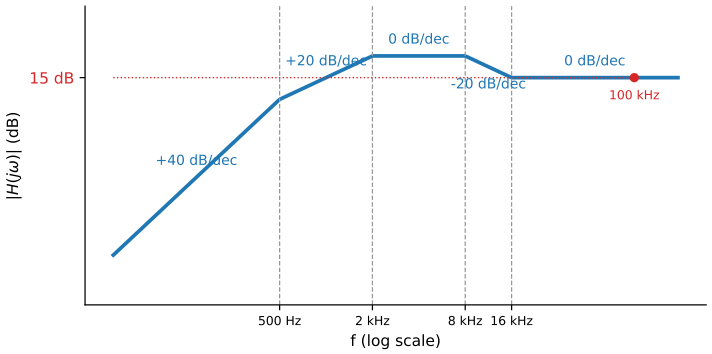
(phase: $\theta + \angle H = \frac{\pi}{4} + (0.464) = 1.25\,\text{rad.}$)

$$\boxed{V_o(t) = 0.894\ \sin\left(2\pi \cdot 2000\,t + 1.25\right)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 1\text{k}} \quad (\text{corners: } 1\text{ kHz})$$
$$V_o(t) = 0.894\ \sin\left(2\pi \cdot 2000\,t + 1.25\right)\ \text{V}$$

Solution 1D.13 — Hard



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
500 Hz	$+40$	$+20$	-20	one pole at f_c
2 kHz	$+20$	0	-20	one pole at f_c
8 kHz	0	-20	-20	one pole at f_c
16 kHz	-20	0	$+20$	one zero at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+40$ dB/dec, i.e. $|H| \propto \omega^2$ at low frequency: the numerator contains s^2. As in class, pair each s with one of the *lowest* poles (500 Hz, 2 kHz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 500}\,\frac{s}{s+2\pi\cdot 2\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 8\text{k}}}\left(1+\frac{s}{2\pi\cdot 16\text{k}}\right)$$

Step 4 — Find K from the marked level

The plot marks the level 15 dB at $f = 100$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 500}$	500 Hz	$f > f_c$: ≈ 1	0
$\dfrac{s}{s + 2\pi \cdot 2\text{k}}$	2 kHz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 8\text{k}}}$	8 kHz	$f > f_c$: $\approx \frac{p}{\omega}$	-21.9
$\left(1 + \dfrac{s}{2\pi \cdot 16\text{k}}\right)$	16 kHz	$f > f_c$: $\approx \frac{\omega}{z}$	15.9
sum			$\mathbf{-6.02}$

The dB values of the factors *add* (magnitudes multiply), so

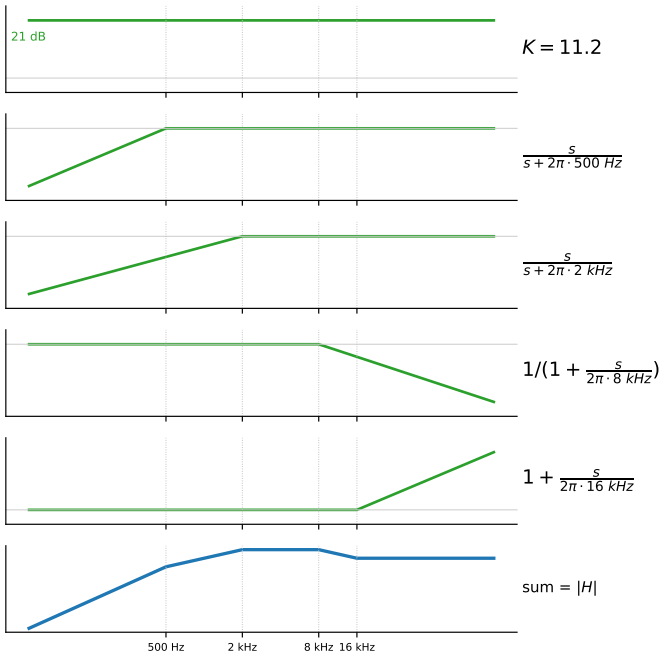
$$20 \log_{10} K = 15 - (-6.02) = 21 \text{ dB} \quad \Rightarrow \quad K = 10^{21/20} = 11.2$$

So

$$\boxed{H(s) = 11.2 \, \frac{s}{s + 2\pi \cdot 500} \, \frac{s}{s + 2\pi \cdot 2\text{k}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 8\text{k}}} \, \left(1 + \frac{s}{2\pi \cdot 16\text{k}}\right)}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 500$ Hz	+40 dB/dec	8.98 dB at end
500 Hz – 2 kHz	+20 dB/dec	8.98 dB at start, 21 dB at end
2 kHz – 8 kHz	0 dB/dec	21 dB at start, 21 dB at end
8 kHz – 16 kHz	−20 dB/dec	21 dB at start, 15 dB at end
$f > 16$ kHz	0 dB/dec	15 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.3\sin(2\pi \cdot 10000\,t)$, so $\omega_{in} = 2\pi \cdot 10000 = 62800$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
11.2	11.25	0
$\dfrac{s}{s+2\pi\cdot 500}$	0.9988	0.05
$\dfrac{s}{s+2\pi\cdot 2\text{k}}$	0.9806	0.197
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 8\text{k}}}$	0.6247	−0.896
$\left(1+\dfrac{s}{2\pi\cdot 16\text{k}}\right)$	1.179	0.559
total	**8.114** (product)	**−0.0901** (sum)

$$|H(j\omega_{in})| = 8.11, \qquad \angle H(j\omega_{in}) = -0.0901 \text{ rad}$$

$$v_{o,ac}(t) = 8.11 \times 0.3\sin(2\pi\cdot 10000\,t - 0.0901) = 2.43\sin(2\pi\cdot 10000\,t - 0.0901)$$

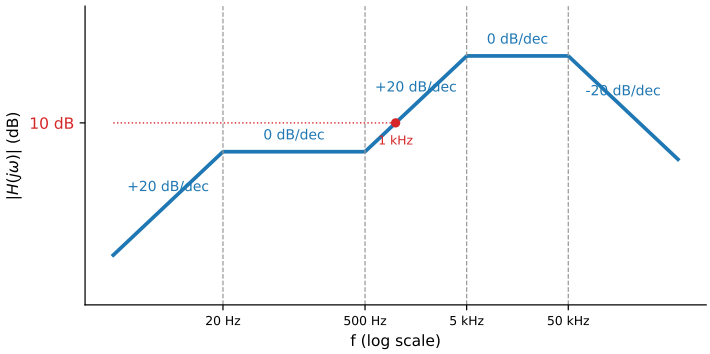
$$\boxed{V_o(t) = 2.43\sin(2\pi\cdot 10000\,t - 0.0901)\ \text{V}}$$

Answer

$H(s) = 11.2\,\dfrac{s}{s+2\pi\cdot 500}\,\dfrac{s}{s+2\pi\cdot 2\text{k}}\,\dfrac{1}{1+\dfrac{s}{2\pi\cdot 8\text{k}}}\left(1+\dfrac{s}{2\pi\cdot 16\text{k}}\right)$ (corners: 500 Hz, 2 kHz, 8 kHz, 16 kHz)

$V_o(t) = 2.43\sin(2\pi\cdot 10000\,t - 0.0901)$ V

Solution 1D.14 Hard



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
20 Hz	$+20$	0	-20	one pole at f_c
500 Hz	0	$+20$	$+20$	one zero at f_c
5 kHz	$+20$	0	-20	one pole at f_c
50 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (20 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 20}\left(1+\frac{s}{2\pi\cdot 500}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 10 dB at $f = 1$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 1$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 20}$	20 Hz	$f > f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 500}\right)$	500 Hz	$f > f_c$: $\approx \frac{\omega}{z}$	6.02
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}$	5 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}$	50 kHz	$f < f_c$: ≈ 1	0
sum			**6.02**

The dB values of the factors *add* (magnitudes multiply), so

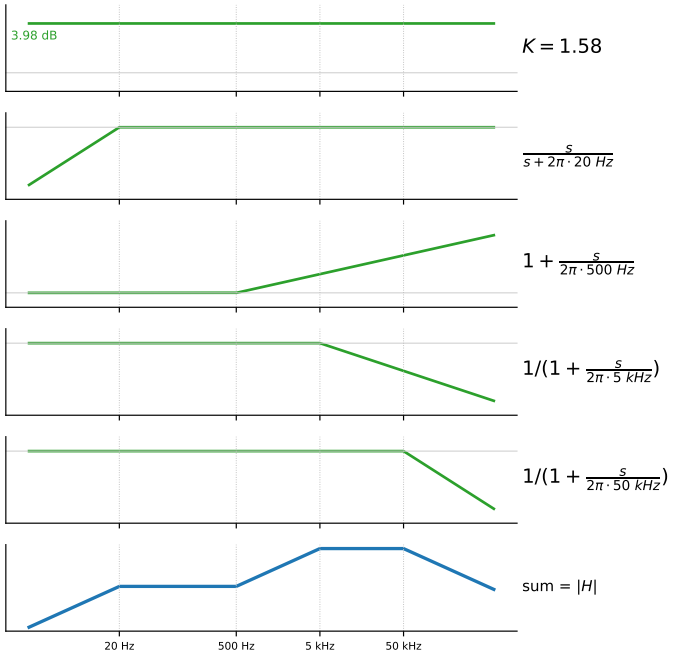
$$20 \log_{10} K = 10 - (6.02) = 3.98 \text{ dB} \quad \Rightarrow \quad K = 10^{3.98/20} = 1.58$$

So

$$\boxed{H(s) = 1.58 \, \frac{s}{s + 2\pi \cdot 20} \left(1 + \frac{s}{2\pi \cdot 500}\right) \frac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 20$ Hz	+20 dB/dec	3.98 dB at end
20 Hz – 500 Hz	0 dB/dec	3.98 dB at start, 3.98 dB at end
500 Hz – 5 kHz	+20 dB/dec	3.98 dB at start, 24 dB at end
5 kHz – 50 kHz	0 dB/dec	24 dB at start, 24 dB at end
$f > 50$ kHz	−20 dB/dec	24 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.2\sin\left(2\pi\cdot 2000\,t + \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1.58	1.581	0
$\dfrac{s}{s+2\pi\cdot 20}$	1	0.01
$\left(1+\dfrac{s}{2\pi\cdot 500}\right)$	4.123	1.33
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$	0.9285	−0.381
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$	0.9992	−0.04
total	**6.048** (product)	**0.915** (sum)

$$|H(j\omega_{in})| = 6.05, \qquad \angle H(j\omega_{in}) = 0.915 \text{ rad}$$
$$v_{o,ac}(t) = 6.05\times 0.2\,\sin\left(2\pi\cdot 2000\,t + 1.96\right) = 1.21\,\sin\left(2\pi\cdot 2000\,t + 1.96\right)$$

(phase: $\theta + \angle H = \frac{\pi}{3} + (0.915) = 1.96\,\text{rad}$.)

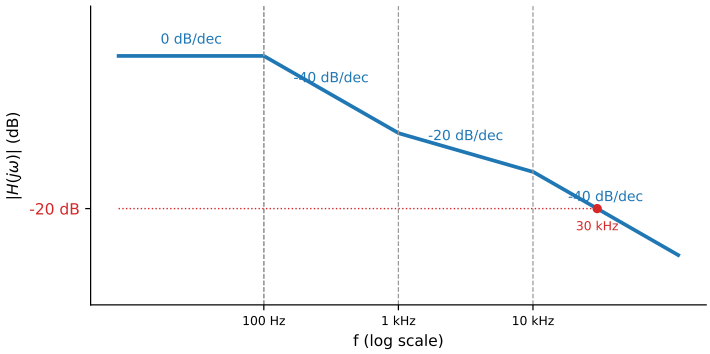
$$\boxed{V_o(t) = 1.21\,\sin\left(2\pi\cdot 2000\,t + 1.96\right)\ \text{V}}$$

Answer

$H(s) = 1.58\,\dfrac{s}{s+2\pi\cdot 20}\left(1+\dfrac{s}{2\pi\cdot 500}\right)\dfrac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}\dfrac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$ (corners: 20 Hz, 500 Hz, 5 kHz, 50 kHz)

$V_o(t) = 1.21\,\sin\left(2\pi\cdot 2000\,t + 1.96\right)$ V

Solution 1D.15 Hard



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	0	-40	-40	two poles at f_c
1 kHz	-40	-20	$+20$	one zero at f_c
10 kHz	-20	-40	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level -20 dB at $f = 30$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 30$ kHz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	100 Hz	$f > f_c$: $\approx \frac{p}{\omega}$	-49.5
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	100 Hz	$f > f_c$: $\approx \frac{p}{\omega}$	-49.5
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1 kHz	$f > f_c$: $\approx \frac{\omega}{z}$	29.5
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	10 kHz	$f > f_c$: $\approx \frac{p}{\omega}$	-9.54
sum			$\mathbf{-79.1}$

The dB values of the factors *add* (magnitudes multiply), so

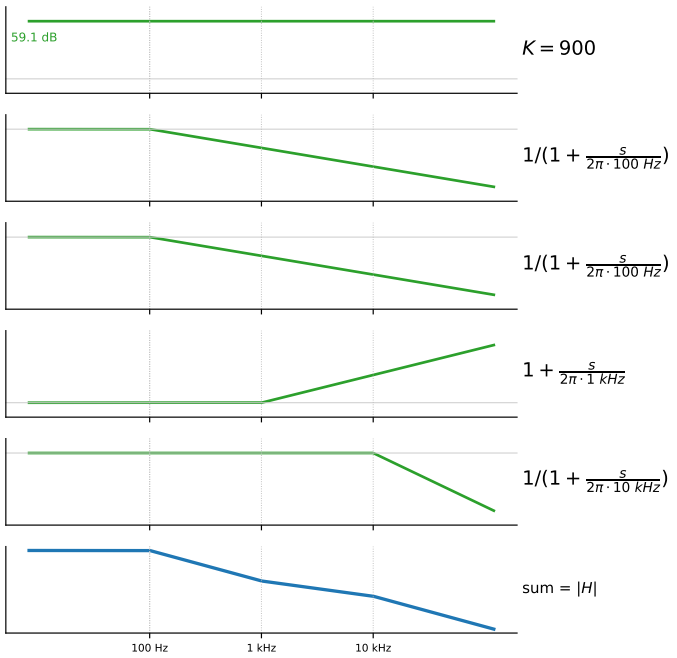
$$20\log_{10} K = -20 - (-79.1) = 59.1 \text{ dB} \quad \Rightarrow \quad K = 10^{59.1/20} = 900$$

So

$$\boxed{H(s) = 900\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\,\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\,\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 100$ Hz	0 dB/dec	59.1 dB at end
100 Hz – 1 kHz	−40 dB/dec	59.1 dB at start, 19.1 dB at end
1 kHz – 10 kHz	−20 dB/dec	19.1 dB at start, -0.915 dB at end
$f > 10$ kHz	−40 dB/dec	-0.915 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 300\,t)$, so $\omega_{in} = 2\pi \cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
900	900	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100}}$	0.3162	−1.25
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100}}$	0.3162	−1.25
$\left(1+\dfrac{s}{2\pi \cdot 1\text{k}}\right)$	1.044	0.291
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}}$	0.9996	−0.03
total	**93.92** (product)	**−2.24** (sum)

$$|H(j\omega_{in})| = 93.9, \qquad \angle H(j\omega_{in}) = -2.24 \text{ rad}$$
$$v_{o,ac}(t) = 93.9 \times 0.1 \sin(2\pi \cdot 300\,t - 2.24) = 9.39 \sin(2\pi \cdot 300\,t - 2.24)$$

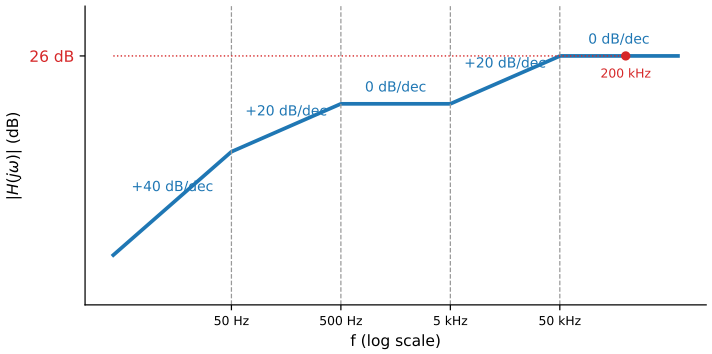
$$\boxed{V_o(t) = 9.39 \sin(2\pi \cdot 300\,t - 2.24)\ \text{V}}$$

Answer

$H(s) = 900\,\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100}}\,\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100}}\left(1+\dfrac{s}{2\pi \cdot 1\text{k}}\right)\dfrac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}}$ (corners: 100 Hz, 100 Hz, 1 kHz, 10 kHz)

$V_o(t) = 9.39 \sin(2\pi \cdot 300\,t - 2.24)$ V

Solution 1D.16 Hard



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
50 Hz	$+40$	$+20$	-20	one pole at f_c
500 Hz	$+20$	0	-20	one pole at f_c
5 kHz	0	$+20$	$+20$	one zero at f_c
50 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+40$ dB/dec, i.e. $|H| \propto \omega^2$ at low frequency: the numerator contains s^2. As in class, pair each s with one of the *lowest* poles (50 Hz, 500 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 50}\,\frac{s}{s+2\pi\cdot 500}\left(1+\frac{s}{2\pi\cdot 5\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 26 dB at $f = 200$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 200$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 50}$	50 Hz	$f > f_c$: ≈ 1	0
$\dfrac{s}{s + 2\pi \cdot 500}$	500 Hz	$f > f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 5\text{k}}\right)$	5 kHz	$f > f_c$: $\approx \frac{\omega}{z}$	32
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}$	50 kHz	$f > f_c$: $\approx \frac{p}{\omega}$	-12
sum			**20**

The dB values of the factors *add* (magnitudes multiply), so

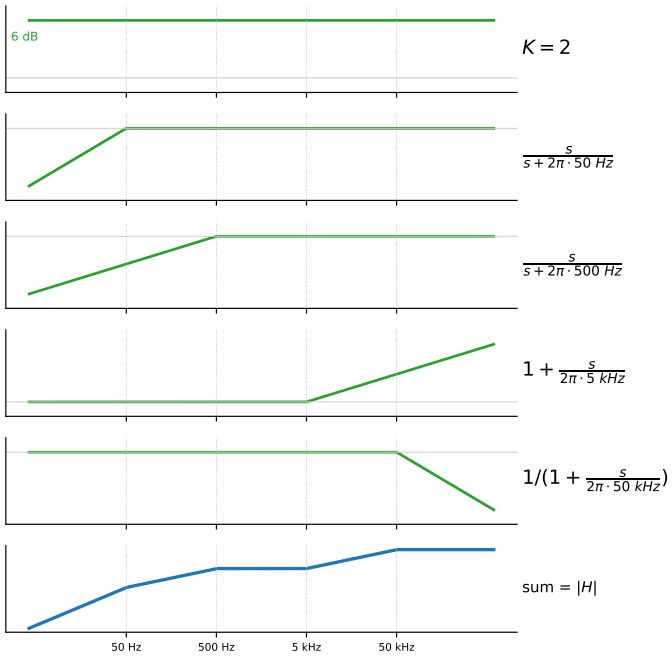
$$20 \log_{10} K = 26 - (20) = 6 \text{ dB} \quad \Rightarrow \quad K = 10^{6/20} = 2$$

So

$$\boxed{H(s) = 2 \, \frac{s}{s + 2\pi \cdot 50} \, \frac{s}{s + 2\pi \cdot 500} \left(1 + \frac{s}{2\pi \cdot 5\text{k}}\right) \frac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 50$ Hz	+40 dB/dec	-14 dB at end
50 Hz – 500 Hz	+20 dB/dec	-14 dB at start, 6 dB at end
500 Hz – 5 kHz	0 dB/dec	6 dB at start, 6 dB at end
5 kHz – 50 kHz	+20 dB/dec	6 dB at start, 26 dB at end
$f > 50$ kHz	0 dB/dec	26 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.05\cos\left(2\pi \cdot 1000\,t - \frac{\pi}{6}\right)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
2	1.995	0
$\dfrac{s}{s+2\pi\cdot 50}$	0.9988	0.05
$\dfrac{s}{s+2\pi\cdot 500}$	0.8944	0.464
$\left(1+\dfrac{s}{2\pi\cdot 5\text{k}}\right)$	1.02	0.197
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$	0.9998	−0.02
total	**1.817** (product)	**0.691** (sum)

$$|H(j\omega_{in})| = 1.82, \qquad \angle H(j\omega_{in}) = 0.691 \text{ rad}$$
$$v_{o,ac}(t) = 1.82 \times 0.05\,\cos\left(2\pi\cdot 1000\,t + 0.167\right) = 0.0909\,\cos\left(2\pi\cdot 1000\,t + 0.167\right)$$

(phase: $\theta + \angle H = -\frac{\pi}{6} + (0.691) = 0.167\,\text{rad}$.)

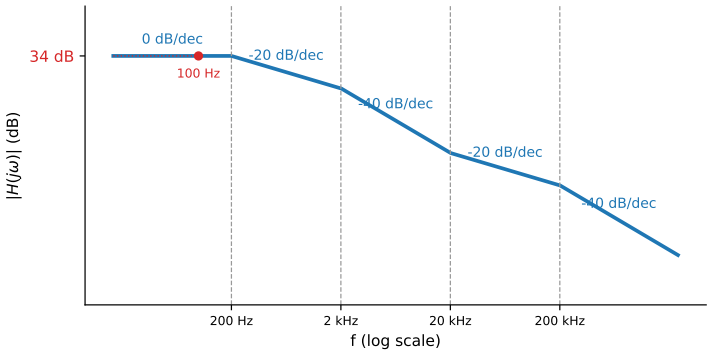
$$\boxed{V_o(t) = 0.0909\,\cos\left(2\pi\cdot 1000\,t + 0.167\right)\ \text{V}}$$

Answer

$H(s) = 2\,\dfrac{s}{s+2\pi\cdot 50}\,\dfrac{s}{s+2\pi\cdot 500}\left(1+\dfrac{s}{2\pi\cdot 5\text{k}}\right)\dfrac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$ (corners: 50 Hz, 500 Hz, 5 kHz, 50 kHz)

$V_o(t) = 0.0909\,\cos\left(2\pi\cdot 1000\,t + 0.167\right)$ V

Solution 1D.17 **Hard**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
200 Hz	0	-20	-20	one pole at f_c
2 kHz	-20	-40	-20	one pole at f_c
20 kHz	-40	-20	$+20$	one zero at f_c
200 kHz	-20	-40	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 200}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\left(1+\frac{s}{2\pi\cdot 20\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 34 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200}}$	200 Hz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$	2 kHz	$f < f_c$: ≈ 1	0
$\left(1+\dfrac{s}{2\pi\cdot 20\text{k}}\right)$	20 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}$	200 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

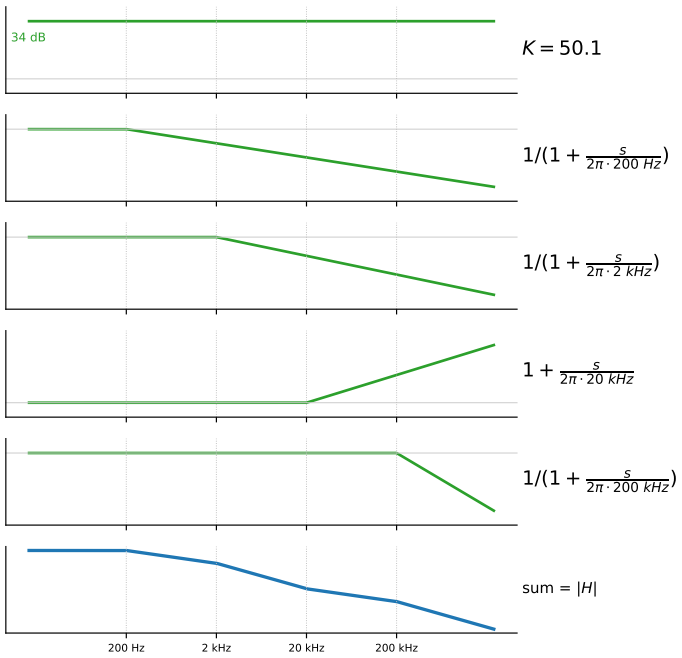
$$20\log_{10} K = 34 - (0) = 34 \text{ dB} \quad \Rightarrow \quad K = 10^{34/20} = 50.1$$

So

$$\boxed{H(s) = 50.1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 200}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\,\left(1+\frac{s}{2\pi\cdot 20\text{k}}\right)\,\frac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 200$ Hz	0 dB/dec	34 dB at end
200 Hz – 2 kHz	-20 dB/dec	34 dB at start, 14 dB at end
2 kHz – 20 kHz	-40 dB/dec	14 dB at start, -26 dB at end
20 kHz – 200 kHz	-20 dB/dec	-26 dB at start, -46 dB at end
$f > 200$ kHz	-40 dB/dec	-46 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.02\sin\left(2\pi\cdot 5000\,t + \frac{\pi}{2}\right)$, so $\omega_{in} = 2\pi\cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
50.1	50.12	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200}}$	0.03997	-1.53
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$	0.3714	-1.19
$\left(1+\dfrac{s}{2\pi\cdot 20\text{k}}\right)$	1.031	0.245
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}$	0.9997	-0.025
total	**0.7666** (product)	$\mathbf{-2.5}$ (sum)

$$|H(j\omega_{in})| = 0.767, \qquad \angle H(j\omega_{in}) = -2.5 \text{ rad}$$
$$v_{o,ac}(t) = 0.767\times 0.02\sin\left(2\pi\cdot 5000\,t - 0.93\right) = 0.0153\sin\left(2\pi\cdot 5000\,t - 0.93\right)$$

(phase: $\theta + \angle H = \frac{\pi}{2} + (-2.5) = -0.93\,\text{rad}$.)

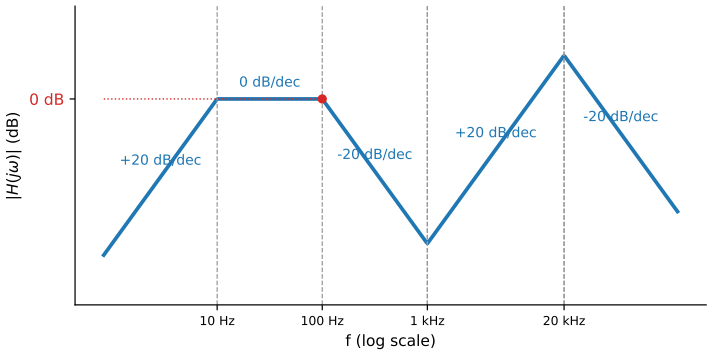
$$\boxed{V_o(t) = 0.0153\sin\left(2\pi\cdot 5000\,t - 0.93\right)\ \text{V}}$$

Answer

$H(s) = 50.1\,\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200}}\,\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\left(1+\dfrac{s}{2\pi\cdot 20\text{k}}\right)\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}$ (corners: 200 Hz, 2 kHz, 20 kHz, 200 kHz)

$V_o(t) = 0.0153\sin\left(2\pi\cdot 5000\,t - 0.93\right)$ V

Solution 1D.18 Hard



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
10 Hz	$+20$	0	-20	one pole at f_c
100 Hz	0	-20	-20	one pole at f_c
1 kHz	-20	$+20$	$+40$	two zeros at f_c
20 kHz	$+20$	-20	-40	two poles at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (10 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 10}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 0 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 10}$	10 Hz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 100}}$	100 Hz	$f < f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}$	20 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}$	20 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

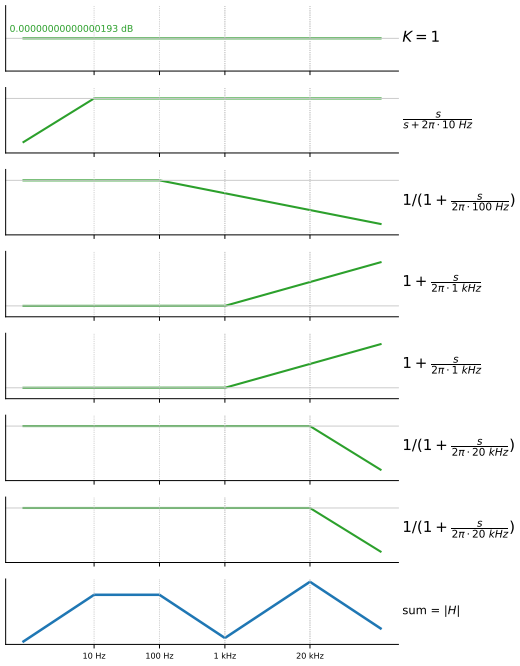
$$20 \log_{10} K = 0 - (0) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1 \, \frac{s}{s + 2\pi \cdot 10} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 100}} \, \left(1 + \frac{s}{2\pi \cdot 1\text{k}}\right) \left(1 + \frac{s}{2\pi \cdot 1\text{k}}\right) \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}}$$

Step 5 — Check

Rebuild the plot from the factors: each factor's straight-line plot (green) is added to get $|H|$ (blue). It must reproduce every slope of the given plot and the marked level.



frequency range	slope	asymptote level
$f < 10$ Hz	+20 dB/dec	0 dB at end
10 Hz – 100 Hz	0 dB/dec	0 dB at start, 0 dB at end
100 Hz – 1 kHz	−20 dB/dec	0 dB at start, -20 dB at end
1 kHz – 20 kHz	+20 dB/dec	-20 dB at start, 6.02 dB at end
$f > 20$ kHz	−20 dB/dec	6.02 dB at start

Step 6 — Steady-state output

The input is $V_i(t) = 0.5 \sin(2\pi \cdot 3000\, t)$, so $\omega_{in} = 2\pi \cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5; use the *exact* factors here, not the straight lines): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 10}$	1	0.00333
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	0.03331	-1.54
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	3.162	1.25
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	3.162	1.25
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9889	-0.149
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9889	-0.149
total	**0.3258** (product)	**0.666** (sum)

$$|H(j\omega_{in})| = 0.326, \qquad \angle H(j\omega_{in}) = 0.666 \text{ rad}$$
$$v_{o,ac}(t) = 0.326 \times 0.5 \sin\left(2\pi\cdot 3000\,t + 0.666\right) = 0.163 \sin\left(2\pi\cdot 3000\,t + 0.666\right)$$

$$\boxed{V_o(t) = 0.163 \sin\left(2\pi\cdot 3000\,t + 0.666\right) \text{ V}}$$

Answer

$H(s) = 1 \dfrac{s}{s+2\pi\cdot 10} \dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}} \left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right) \left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right) \dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}} \dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$ (corners: 10 Hz, 100 Hz, 1 kHz, 1 kHz, 20 kHz, 20 kHz)

$V_o(t) = 0.163 \sin\left(2\pi\cdot 3000\,t + 0.666\right)$ V